

Enzymatically hydrolyzed and TEMPO-oxidized cellulose nanofibers for the production of nanopapers: morphological, optical, thermal and mechanical properties

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Abstract In the present study, CNF prepared by TEMPO-mediated oxidation and enzymatic hydrolysis were used for the production of nanopapers using a papermaking-like route. Nanopapers were characterized in terms of tensile, thermal, optical and morphological properties. Those prepared from enzymatically hydrolyzed CNF were found to be weaker at tensile than those resulting from TEMPO-mediated oxidation, but with similar level of stiffness. Enzymatically obtained CNF presented lower transparency due to their higher diameter and lower fibrillation yield. Moreover, TEMPO-oxidized CNF presented lower onset of the thermal degradation temperature (230 °C) due to the presence of carboxylic groups. Overall, the influence of increasing the amount of enzyme during enzymatic hydrolysis and the amount of sodium hypochlorite during TEMPO-mediated oxidation was assessed as function of the ultimate properties of nanopapers.

Keywords Cellulose nanofibers · Nanopaper · Enzymatic hydrolysis · TEMPO-mediated oxidation

Introduction

In the last century, many efforts have been paid to sustainable, green and environmental friendly materials for a wide range of applications. Among these materials, cellulose deserves special attention due to its abundance, renewability, lightweight, recyclability, biodegradability and many other interesting properties and characteristics (Saito et al. 2007). Cellulose is the most common and abundant organic compound and biopolymer on Earth, and it is present in many living species. Among these species, the main sources of cellulose worldwide are trees, annual plants and agricultural residues, which, unfortunately, have a finite supplying capacity. In addition, forests and croplands are continuously competing with the need of using such lands for agricultural purposes. In this sense, bearing in mind that the supplying capacity is such limited, cellulose fibers extraction processes must be improved and optimized to obtain the maximum efficiency in terms of natural resources exploitation (Hubbe 2014).

Cellulose nanofibers (CNF) have become a topic of increasing interest with promising applications in a broad field of innovative nanostructured materials. CNF are the smallest structural unit of plant fiber, consisting of a bundle of stretched cellulose chain molecules with long, flexible and entangled cellulose nanofibers which alternate crystalline and amorphous domains. In the same way than common cellulose

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fibers, nanocellulose extends the potential application to papermaking (Eriksen et al. 2008), nanopapers (Sehaqui et al. 2010), biomedicine (Jorfi and Foster 2015) and nanocomposites (Spence et al. 2011), among others. Furthermore, CNF have greater mechanical properties than common fibers, contributing to strongly enhance the mechanical performance in comparison with cellulosic fibers. In this sense and in the aforementioned frame of increasing the efficiency of natural resources exploitation, the higher performance of CNF confer them the feasibility to be used in common applications, but using lower amounts, contributing thus both to economy and environment (Hubbe 2014).

CNF can be produced by high shear mechanical disintegration process using high-pressure homogenizers, grinders or microfluidizers. To facilitate the disintegration process and increase the yield in cellulose nanofibrils, pretreatment such as TEMPO-mediated oxidation, enzymatic hydrolysis, carboxymethylation and mechanical refining, is widely used (Delgado-Aguilar et al. 2015a). Depending on the selected pre-treatment, the obtained CNF will present specific properties, as well as environmental impact and production costs. In that way, several authors have reported the need of producing tailor-made CNF for each application, always pursuing a well-balanced triangle between properties, costs and environmental impact (Henriksson et al. 2007). Nonetheless, several efforts must to be already paid to further understand production methods and properties of cellulose nanofibers (Naderi 2017).

One of the most promising CNF-based products is nanopaper, which consists of a membrane of a 3-D structured CNF network. Mimicking papermaking process, nanopapers can be produced by filtering a CNF suspension, leading to a fully cellulose-based membrane with high mechanical properties (Henriksson et al. 2008), being more environmentally friendly than plastic films and made of renewable and biodegradable raw materials. Moreover, when TEMPO-oxidized CNF are used, nanopapers present high transparency that able them to be used as substrate for optoelectronic devices, loudspeaker membranes, photovoltaic devices and transistors, among others (Olsson et al. 2010; Zhu et al. 2014). However, such kind of CNF have a great capacity to absorb water, increasing the filtration time and, thus, increasing the energy consumption for their

production. In this sense, several strategies must be implemented to develop nanopapers from CNF with lower capacity to retain water without decreasing their mechanical properties. Thus, the present work aims to compare the mechanical, morphological, optical and thermal properties of nanopapers made both from TEMPO-oxidized and enzymatically hydrolyzed CNF.

Materials and methods

Materials

Bleached kraft hardwood (*Eucalyptus globulus*) pulp (BKHP) was kindly provided by Torraspapel S.A. (Spain), originally with 16 μm of diameter and 700 μm of length, according to the supplier. The enzymatic hydrolysis was performed using the commercial enzyme cocktail Novozym 476, kindly provided by Novozymes A/S (Denmark), that contains 2% of endo- β -1,4-glucanases with an activity factor of 4500 CNF-CA/g of cellulose (tested over a CMC substrate). All the reagents required for conducting both enzymatic hydrolysis and TEMPO-catalyzed oxidation, as well as for CNF characterization, were supplied by Sigma Aldrich (Spain).

Cellulose nanofibers production

TEMPO-mediated oxidation was performed maintaining pH at 10, according to the methodology reported by Saito et al. (2007). For the production of fifteen grams of CNF, 1.5 g NaBr and 0.24 g TEMPO (2,2,6,6-tetramethylpiperidine-1-oxyl radical) were diluted into 1.2 L distilled water. When both TEMPO and NaBr were completely dissolved in water, 15 g of BKHP fibers, prior disintegration, were added to the solution, adding extra water until reaching a total volume of 1.5 L. Keeping the suspension under gentle stirring, NaClO was gradually dropped in. The amount of NaClO was ranged between 5 and 15 mmols/g. Once all NaClO was added, NaOH at 0.1 M was dosed until achieving a constant pH of 10. The oxidized suspensions then were washed with distilled water and kept at 4 °C.

Enzymatic hydrolysis was carried out according to the methodology reported by (Henriksson et al. 2007). However, several parameters were changed according

to the optimization carried out by Tarrés et al. (2016). BKHP was dispersed at 1.5 wt% in water in a laboratory pulper for 30 min at 3000 rpm. Then, the fibers were filtered until 10 wt% consistency and refined in a PFI mill for 4000 revolutions. This process was carried out to swell the fibers and thus to promote the activity of enzymes. Briefly, refined fibers were suspended in water again (until reaching a pulp consistency of 5 wt%), and 0.1 N HCl was dropped until achieving a pH of 5. Then, the suspension was heated until 50 °C under constant stirring to avoid temperature gradients. At this step, the enzyme cocktail was dropped into the suspension and stirred for 4 h. The enzymatic process was stopped by heating the suspension to 80 °C for 15 min, where the enzyme activity is strongly decreased (de Marco and Felix 2007). Enzyme dosage was ranged between 80 and 320 g/Tn. The enzymatically hydrolyzed pulp then was washed with distilled water and kept at 4 °C.

The treated suspensions (1 wt% consistency) were gradually homogenized in a PANDA Plus laboratory homogenizer (Gea Niro Soavi, Italy) following the sequence of three passes at 300 bar, three passes at 600 bar and three passes at 900 bar. All the suspensions followed the abovementioned sequence, making the difference only on the type of pretreatment. Moreover, the pressure was increased gradually to avoid clogging in the pressure chambers.

The produced CNF received the following nomenclature: CNF-Treatment/Dosage. TEMPO-oxidized CNF received the following codes: CNF-T/5, CNF-T/10 and CNF-T/15, being the oxidation performed at 5, 10 and 15 mmols/g of NaClO, respectively. Enzymatically hydrolyzed CNF were labeled as follows: CNF-E/80, CNF-E/160, CNF-E/240 and CNF-E/320.

Nanopapers production

Nanopapers were prepared mimicking the papermaking process at laboratory scale. In all cases, 152 g of the CNF gel at 1 wt% was diluted with water until reaching a consistency of 0.2 wt% (760 g, wet weight). Then, CNF were dispersed by stirring the suspension at 20,000 rpm for 2 min by means of Ultraturrax T25 (IKA, Germany). Then, the suspension was poured in a modified sheet former equipped with a 0.22 µm nitrocellulose membrane at the bottom. Suspensions were then vacuum-dewatered

until a gel-like cake of CNF was observed at the top of the membrane. The obtained membranes were then vacuum dried for 15 min. The whole process is summarized in Fig. 1. All the samples were conditioned at 23 °C and 50% of relative humidity for 24 h for testing.

Cellulose nanofibers characterization

The characterization of CNF was performed according to previous works (Delgado-Aguilar et al. 2015a). Cationic demand was determined by colloidal titration using a Mutek PCD04 charger analyzer from BTG, S.L. (UK). This titration was performed through surface adsorption of poly-DADMAC and the excess was titrated with PES-Na and anionic standard polymer, both from BTG. The molecular weight (MW) of the poly-DADMAC was 107 kDa.

The carboxyl content of the oxidised cellulose was determined using conductometric titration, as described elsewhere (Besbes et al. 2011).

Specific surface and diameter were estimated from cationic demand and carboxyl content, as described by Espinosa et al. (2016). Briefly, it was considered that the interaction between the CNF surface and the added cationic polymer (poly-DADMAC) occurred through two different mechanisms: on the one hand, part of the polymer got retained by ionic interaction between carboxylic groups from CNF and the polymer thereof. On the other, the rest of the consumed poly-DADMAC during cationic demand determination was assumed to be retained by hydrogen bonding and Van der Waals forces. This can be assumed due to the high molecular weight of the poly-DADMAC that was used. Indeed, low-MW polymers could penetrate into the cellulose fiber cell walls, while high-MW polymers cannot (Lizundia et al. 2016).

Moreover, the estimation of the specific surface was accomplished taking into account two assumptions: (1) the surface adsorption of poly-DADMAC took place in the form of a monolayer and (2) the poly-DADMAC chain has a cylindrical geometry. The surface of poly-DADMAC was initially estimated by calculating the surface of the monomer and its polymerization degree, taking into account the bond distances and assuming a cylindrical monomer ($535.87 \text{ nm}^2/\text{g}$, being equal to $4.87 \cdot 10^{17} \text{ nm}^2/\mu\text{eq-g}$).

Moreover, using the values obtained for cationic demand and the carboxyl content it was possible to

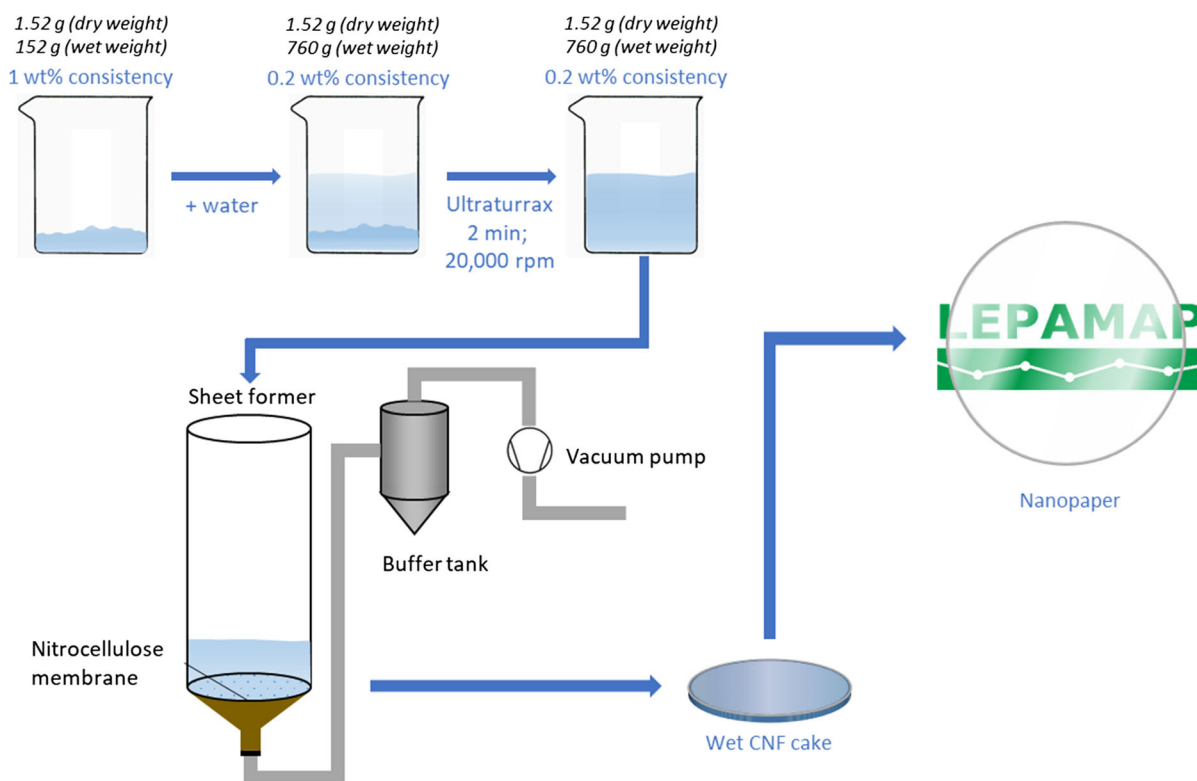


Fig. 1 Nanopaper manufacturing process

calculate the CNF's theoretical surface area by the following equation:

$$\sigma_{CNF} = (CD - CC) \cdot S_{Poly-DADMAC}$$

where CD is cationic demand, CC is carboxyl content (both expressed in $\mu\text{eq-g/g}$) and $S_{Poly-DADMAC}$ is the specific surface of Poly-DADMAC, in $\text{nm}^2/\mu\text{eq-g}$.

Finally, assuming that nanofibers are perfect cylinders and that cellulose density is 1.5 g/cm^3 , diameter was estimated.

The yield of nanofibrillation was determined by centrifuging (rotational speed of 4000 rpm; 3100 G-force) an aqueous suspension of CNF (50 mL; 0.05 wt%) in a vessel equipped with a nitrocellulose membrane (pore size of $0.22 \mu\text{m}$) at the bottom. This concentration was selected to be in the range of the critical networking concentration (Naderi et al. 2014). The retained solids were weighted and referred to the total amount of CNF added to the vessel, obtaining the non-fibrillated percentage. The degree of polymerization (DP) of CNF was determined from intrinsic viscosity measurements, according to UNE 57-039-92. The viscosimetric average molecular weight was

calculated from the equation: $g = KMa$, where g is the intrinsic viscosity, $K = 2.28$ and $a = 0.76$ (Henriksson et al. 2008). Transmittance measurements were performed on CNF suspensions with 0.1% solid content. The sample was introduced into quartz cuvettes and the transmittance measured with a UV-Vis Shimadzu spectrophotometer UV-160A set in the range between 400 and 800 nm. Distilled water was used as reference.

Nanopapers characterization

Density and porosity were calculated from the basis weight and thickness of the obtained nanopapers. Nanopapers were mechanically tested in an Instron universal testing machine provided with a 2.5 kN load cell. The gap between clamps was set at 150 mm and preload was 0.1 N with a cross-head testing velocity of 15 mm/min. The testing specimens were cut down to stripes of 180 mm length and 15 mm width.

Transmittance measurements were performed on nanopapers, introducing the samples in the UV-Vis spectrophotometer without cuvettes. The wavelength

was ranged between 400 and 800 nm, using air as reference.

Water contact angles on nanopapers surface were measured using a DSSA24 drop-shape analyzer from Krüss GmbH (Germany) with the Krüss Advance Software. The measurements were performed at room temperature with a frequency of two measurements per second. The total testing time was 30 s for each nanopaper.

Crystallinity degree was calculated from an X-Ray Diffraction (XRD) profile. XRD measurements were performed on small pieces of nanopaper. Cu K α radiation was generated with a Bruker D8 Advance diffractometer. Sweeps of 5°–50° 2 θ were made with a step size of 0.05 s and the step time measurement was 10 s. The crystallinity was calculated using the diffraction intensities of the crystalline structure and that of the amorphous fraction according to the method reported by Segal et al. (1959):

$$CrI (\%) = \frac{I_H - I_L}{I_H} \cdot 100$$

The morphology of the surface of nanopapers was observed by Field Emission Scanning Electron Microscopy (ZEISS SUPRA400) at an accelerating voltage of 2 kV. The samples were bound to the metal holder using carbon tape and coated with a thin layer of gold.

The thermal behaviour of the nanopapers was evaluated to analyse the thermal stability of fibers, using about 10 mg solid samples and a thermogravimetric analyser Perkin-Elmer Pyris from 20 to 600 °C at a heating rate of 10 °C/min under oxygen condition.

Results and discussion

Cellulose nanofibers characterization

CNF characterization is shown in Table 1.

Those CNF obtained by TEMPO-mediated oxidation showed increasing carboxyl content as the amount of sodium hypochlorite during oxidation was increased. This effect is completely understandable, taking into account that TEMPO/NaBr/NaClO oxidation in water at basic pH selectively oxidizes C6 primary hydroxyls of cellulose to C6 carboxylate groups (Isogai et al. 2011). This increasing carboxyl content led to higher cationic demand of the CNF

suspensions due to the increase on the specific surface thereof. Carboxyl groups, either in its acidic or salt form, have greater volume than hydroxyl groups. Moreover, they impart electrostatic charges between fibers, promoting fibrillation during the homogenization stage. On the other side, enzymatically hydrolyzed CNF presented a constant CC as the amount of enzyme was increased, being the same that for the neat pulp (not shown in the table). Nonetheless, cationic demand was progressively increased, meaning that specific surface was also increased. Hence, Table 1 shows the estimation of the specific surface and average diameters (assuming cylindrical-shaped CNF) of both TEMPO-oxidized and enzymatically hydrolyzed CNF. As reflected, diameter was decreased for increasing amounts of NaClO during TEMPO-mediated oxidation. On the other hand, as the enzyme dosage was increased during enzymatic hydrolysis, no clear effect on CNF diameter was observed from 160 to 320 g/Tn.

The relationship between DP and CNF diameters is interesting to understand the mechanism of each treatment. As reflected in Table 1, in the case of TEMPO-oxidized CNF the obtained DP went from 490 to 197, for 5 to 15 mmols, respectively. On the other hand, those CNF obtained by enzymatic hydrolysis presented slightly lower DP, ranging from 412 to 309 for 80 to 320 g/Tn, respectively. It is true that DP measurement using cooper ethylenediamine (CED) as cellulose solvent may decrease the chain length and, thus, the DP due to β -elimination of adjacent glycoside bonds at the C6-aldehyde units. However, since all the measurements were performed following the same methodology, they can be used to compare the obtained CNF, assuming the same effect of CED on all the samples. Considering that there is a direct relationship between the DP and the length of TEMPO-CNF (Shinoda et al. 2012), aspect ratio can be calculated. Although in principle this relationship is only valid for TEMPO-CNF, the obtained DP values for enzymatic CNF were also used for the calculation of their lengths, leading to plausible values.

Aspect ratio brings to the light the main difference between the two treatments used for CNF production in this work. On the one side, TEMPO-mediated oxidation is able to fibrillate fibers separating fiber bundles, leading to thinner fibers, at the same time that cuts cellulose chains and decreases the fiber length. On the other, enzymatic hydrolysis acts in the amorphous

Table 1 Characterization of CNF-T and CNF-E: carboxyl content, cationic demand, transmittance, crystallinity index, degree of polymerization, yield and morphological parameters

Sample	CC ($\mu\text{eq-g/g}$)	CD ($\mu\text{eq-g/g}$)	σ (m^2/g)	d (nm)	T at 600 nm (%)	CrI (%)	DP (–)	Length ^a (nm)	Aspect ratio	Yield (%)
CNF-T/5	793 $\pm$ 29	1174 $\pm$ 86	185.8	14.4	80.1	50	490 $\pm$ 25	1343	93.3	95.6
CNF-T/10	1160 $\pm$ 97	1707 $\pm$ 80	265.6	10.0	81.6	46	229 $\pm$ 12	224	22.5	96.7
CNF-T/15	1526 $\pm$ 103	2239 $\pm$ 91	347.2	7.7	83.4	52	197 $\pm$ 8	87	11.3	97.1
CNF-E/80	41.8 $\pm$ 3.1	187 $\pm$ 11	70.7	37.7	30.2	56	412 $\pm$ 31	1009	26.8	20.5
CNF-E/160	42.1 $\pm$ 4.0	255 $\pm$ 29	103.7	25.7	33.1	64	320 $\pm$ 30	615	23.9	29.4
CNF-E/240	42.4 $\pm$ 3.5	258 $\pm$ 13	105.0	25.4	38.6	53	307 $\pm$ 16	559	22.0	34.0
CNF-E/320	42.1 $\pm$ 2.2	260 $\pm$ 19	106.1	25.1	43.7	57	309 $\pm$ 18	567	22.6	38.9

Carboxyl content (CC); cationic demand (CD); specific surface (σ); diameter (d); transmittance (T at 600 nm); crystallinity index (CrI); degree of polymerization (DP)

^a Calculated from length (nm) = 4.286·DP – 757

regions of cellulose, cutting cellulose chain and reducing their length more than their diameter. This is why CNF with similar length (CNF-T/5 and CNF-E/80) present such different diameter and, thus aspect ratio.

The higher diameter that CNF-E have is reflected in their transmittance at 600 nm wavelength and yield of fibrillation. In this sense, it is expectable that nanopapers prepared with CNF-T will present higher transmittance than those obtained from CNF-E, being more suitable for optical applications where transparency is a requirement.

All the XRD patterns of CNF exhibited the typical Cellulose I crystalline structure, with peaks located at 15°, 16° and 23° corresponding to ($\bar{1}10$), (110) and (200) reflection planes, respectively. The CrI values of the nanopaper from CNF-T and CNF-E were around 50% and in the range of 54–64%, respectively. The higher CrI of nanopaper from CNF-E comes from the higher removal of amorphous regions during the enzymatic treatment.

Physical and morphological characterization of nanopapers

Table 2 shows the evolution of thickness, density, porosity and transmittance both as oxidation degree and enzyme dosage were increased. Basis weight was set at 40 g/m². Variations are attributed to errors while measuring CNF consistency and/or the volume taken for nanopaper formation.

Since thickness was decreased and basis weight was kept constant, density was increased with

increasing amount of NaClO and enzyme for CNF-T and CNF-E, respectively. Apparently, it seems that there is a direct influence of the specific surface of CNF on the nanopaper density. However, comparing CNF-E/320 with CNF-T/15, it can be seen that the obtained nanopapers presented similar density (and porosity) while their specific surfaces were considerably different (see Table 1). Hence, it appears that there is no direct correlation between specific surface and bulk density, although as treatment severities were hardened, bulk density was slightly increased. These bulk density values are significantly higher than those reported in the literature related to nanopapers prepared from CNF-E (Henriksson et al. 2008). These higher densities, and low porosities, are interesting for applications where barrier properties are needed (Österberg et al. 2013). Nonetheless, the achieved densities are slightly lower than already published values (Wu et al. 2012). The obtained nanopapers were produced by filtration and then vacuum dried, fact that limited the shrinkage of CNF and subsequent compaction of CNF. In addition, Wu et al. (2012) used the supernatant of CNF suspension after centrifugation, meaning that the specific surface and aspect ratio of their fibrils was significantly higher to the ones used in this work, enhancing inter-CNF bonding and, thus, promoting compaction as well. However, when the same methodology is followed, similar values can be found (Sehaqui et al. 2010). Transmittance was enhanced as NaClO content was increased for the case of CNF-T. In the same way, increasing amounts of enzyme reduced the opacity of nanopapers made of CNF-E. This increase on transmittance is explained,

Table 2 Physical characterization of the CNF-T and CNF-E nanopapers

Nanopaper	Basis weight (g/m ²)	Thickness (μm)	Density (kg/m ³)	Porosity (%)	Transmittance at 600 nm (%)
CNF-T/5	39.75 ± 2.01	33.4 ± 1.1	1.19	20.7	48.7
CNF-T/10	38.92 ± 1.17	27.8 ± 0.6	1.40	6.7	72.6
CNF-T/15	37.90 ± 1.88	26.5 ± 0.8	1.43	4.7	83.4
CNF-E/80	38.79 ± 0.89	33.7 ± 0.3	1.15	26.5	11.6
CNF-E/160	39.06 ± 1.02	32.3 ± 1.2	1.21	13.2	13.8
CNF-E/240	41.86 ± 2.00	31.2 ± 1.0	1.34	5.4	14.9
CNF-E/320	40.33 ± 2.46	28.3 ± 0.4	1.43	5.0	22.5

on the one hand by the decrease in the amount of non-fibrillated and partially fibrillated fractions responsible for a light-scattering phenomenon, and on the other hand by the decrease in the nanopaper porosity (Besbes et al. 2011).

Wetting of nanopapers was assessed by water contact angle (WCA) measurement and results are reported in Table 3. Those nanopapers made of CNF-T presented WCA of about 85° when drop was deposited, while those prepared with CNF-E had about 10° less. These values are significantly high compared to those reported in the literature, where WCA are usually in the range of 50°–55° (Sun et al. 2015). The high density of surface hydroxyl groups both in cellulose and hemicellulose confer to CNF a huge hydrophilic character and, according to Dankovich and Gray (2011), WCA lower than 40° is expected. The higher values obtained in this work may be explained by two effects: (1) the presence of residual lignin in the form of a thin layer blocking part of the surface hydroxyl groups (Ferrer et al. 2012) and/or (2) the presence of residual extractives. Besides, after 30 s of the drop deposition, WCAs were not excessively decreased, fact that might be explained by the low porosity of the nanopapers, hindering water absorption by capillarity.

Mechanical characterization of nanopapers

Nanopapers were tested at tensile, obtaining the tensile strength (σ_t^{NP}), Young's Modulus (E_t^{NP}) and their elongation at maximum stress (ε_t^{NP}).

Table 4 and Fig. 2 bring to the light the influence of the production method, as well as the severity of the treatment on mechanical and optical properties. In the case of CNF-T, tensile strength was increased from 90.3 to 152.6 MPa as the amount of NaClO was

Table 3 Water contact angle at 0 and 30 s of drop deposition

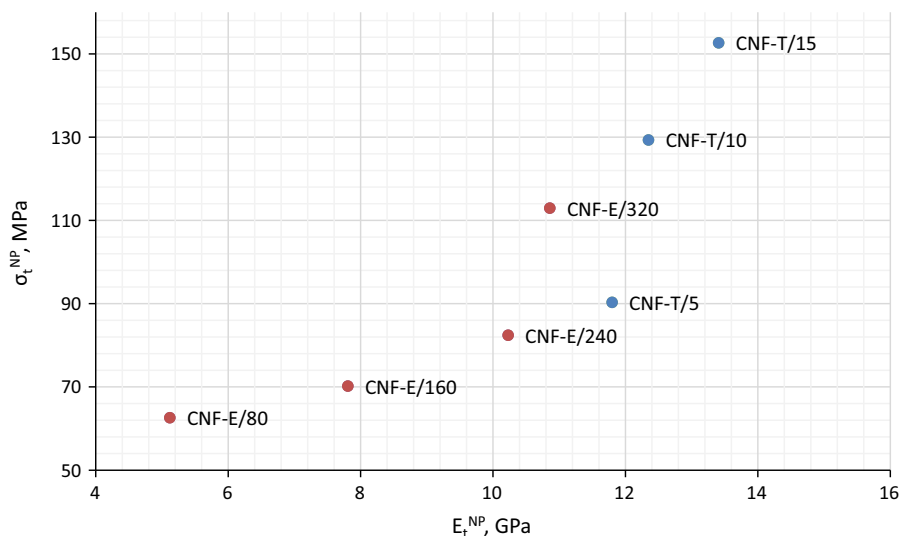
Nanopaper	WCA at 0 s (°)	WCA at 30 s (°)
CNF-T/5	85.5	72.1
CNF-T/10	85.0	71.6
CNF-T/15	85.1	70.7
CNF-E/80	78.3	68.3
CNF-E/160	76.2	67.4
CNF-E/240	75.1	66.0
CNF-E/320	75.0	65.7

increased (from 5 to 15 mmols/g, respectively). On the other hand, the maximum tensile strength that was achieved with enzymatic CNF was 112.9 MPa, being a value in between nanopapers made of CNF-T/5 and CNF-T/10. The differences between the strength of nanopapers from CNF-T and CNF-E are likely due to the differences on the intrinsic properties of each type of CNF. CNF-T presented higher specific surface and yield of fibrillation than CNF-E, fact that provided them higher swelling capacity and, thus, better suitability to form a three-dimension network (Chun et al. 2011). In addition, higher specific surface implies higher density of inter-CNF bonds that, at the same time, provided improved mechanical performance to the resulting CNF-based structures due to higher density. For an equivalent tensile strength, CNF-T presented higher Young's modulus than CNF-E. This could be explained by the fact that CNF-T were significantly shorter than CNF-E, limiting their plastic behavior under stress and, at the same time, their future applications where flexibility was required.

In addition to the strong dependence of the nanopapers properties on the density (Sehaqui et al. 2011), other characteristics are expected to affect

Table 4 Evolution of the tensile properties of the obtained nanopapers

Nanopaper	σ_t^{NP} (MPa)	E_t^{NP} (GPa)	ε_t^{NP} (%)	Specific σ_t^{NP} (MPa cm ³ /g)	Specific E_t^{NP} (GPa cm ³ /g)
CNF-T/5	90.3 ± 3.2	11.8 ± 0.9	1.89 ± 0.09	75.87	9.92
CNF-T/10	129.3 ± 6.4	12.4 ± 1.0	2.38 ± 0.15	96.49	9.22
CNF-T/15	152.6 ± 7.0	13.4 ± 1.1	2.69 ± 0.11	106.72	9.38
CNF-E/80	62.6 ± 2.3	5.1 ± 0.5	1.96 ± 0.08	54.42	4.45
CNF-E/160	70.2 ± 5.0	7.8 ± 0.4	2.03 ± 0.15	57.98	6.45
CNF-E/240	82.4 ± 8.0	10.2 ± 0.9	2.28 ± 0.20	61.49	7.63
CNF-E/320	112.9 ± 9.8	10.9 ± 0.8	2.31 ± 0.21	79.24	7.62

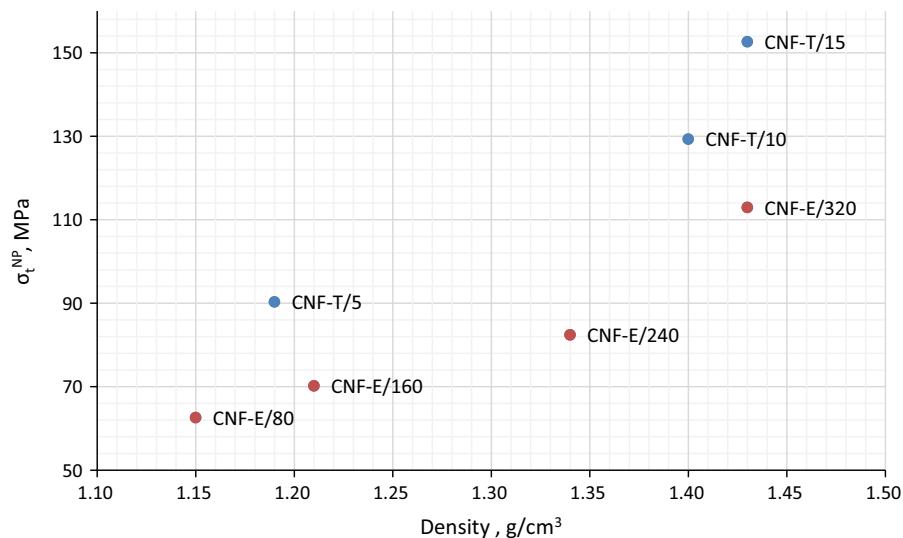
Fig. 2 Correlation between tensile strength and modulus as function of the treatment type and severity

them, including: (1) intrinsic properties of the nanofibers, (2) number of bonds per volume unit and (3) nature of these bonds (Delgado-Aguilar et al. 2015b). In this sense, denser papers will present higher degree of bonds per volume unit and, thus higher mechanical properties. However, as the intrinsic properties of the CNF are much stronger than micro sized cellulose fibers, nanopapers will also present higher mechanical properties compared to common paper. Figure 3 shows the evolution of the tensile strength as density is increased, both for CNF-T and CNF-E.

For a similar value of density (i.e. CNF-T/15 and CNF-E/320), it can be seen that CNF-T nanopaper presented a tensile strength 35% higher than that made from CNF-E. This difference was even higher when comparing CNF-T/10 and CNF-E/240 (57%).

However, the increase on the tensile strength of the nanopapers did not merely result from the increase on the density, but also involved the contribution of the intrinsic mechanical properties of the CNFs, as shown in Table 4. As it is possible to see, as the amount of NaClO (CNF-T) and enzyme (CNF-E) were increased, intrinsic mechanical properties improved. In the case of CNF-T, Young's modulus was increased merely by higher compaction as the oxidation degree was increased, since no significant changes can be observed in the specific E_t^{NP} . On the other hand, for the case of CNF-E, intrinsic CNF modulus was enhanced until an enzyme dosage of 240 g/Tn was achieved, being the same than for CNF-E/320. However, the effect of density was significantly higher on tensile strength than stiffness.

Fig. 3 Evolution of the tensile strength as function of density



Field-emission scanning electron microscopy (FE-SEM)

Transparency is a property that offers a wide range of applications to nanopapers (Hu et al. 2013). This property is driven by the amount of light that is scattered through the nanopaper when light radiates it. Light scattering occurs both at nanopaper surface and along the interfacial area CNF-air, and it depends on the surface roughness and on the morphology of the CNF. In fact, given the strong dependence of light scattering with the particle size ($\propto D^3$) (Althues et al. 2007) then the main parameter governing the transparency of nanopapers would be the width of CNF, the fraction of unfibrillated or partially fibrillated material and the size of the pore of the nanopaper (Fig. 4).

FE-SEM observation was conducted with the purpose of comparing the morphology of CNF as function of the treatment used for their production. In all cases, nanopapers showed randomly oriented and entangled CNF forming a 3-D structure. In the case of those nanopapers obtained from CNF-E, diameters ranging from 5 to 25 nm were observed, while those prepared with CNF-T showed a narrower distribution, being of 5–10 nm. FE-SEM images revealed the huge capacity that CNF have to set up entangled networks through, presumably, inter-CNF hydrogen bonds. However, it is worth to mention that FE-SEM observation at lower magnification (covering a piece of nanopaper of $100 \times 100 \mu\text{m}$, Fig. 5) revealed the presence of fibrils in the micro domain,

arising from the fraction partially fibrillated in both cases (both CNF-T and CNF-E).

The lower transparency of nanopapers made from CNF-E compared to those from CNF-T come mainly from the higher presence of non-nanofibrillated material, which scatters the light more than those fibrils in the nano domain. Recovering the data from Table 1, one can see that yield of fibrillation was ranged between 95.6 and 97.1% in the case of CNF-T and from 20.5 to 38.9% for CNF-E. The yield of fibrillation was increased with the severity of the treatment, as well as density did. Thus, the higher presence of fibrillated material leads to higher specific surface of the CNF suspension, implying more capability to form inter-CNF bonds and, at the same time, form more compacted entangled networks.

Thermogravimetric analysis (TGA)

There are several application fields where nanopapers have potential to be applied and, at the same time, thermal stability is a requirement. In this sense, samples were submitted to thermogravimetric analysis, leading to the curves shown in Fig. 6 with the corresponding peaks (Table 5).

As it can be seen in Fig. 6, thermogravimetric curves experienced three changes during the heating ramp. First of all, at the temperature of about 100°C , absorbed water (moisture) was evaporated, experiencing a weight loss of 7 and 10% for CNF-E and CNF-T, respectively. The higher specific surface and carboxyl

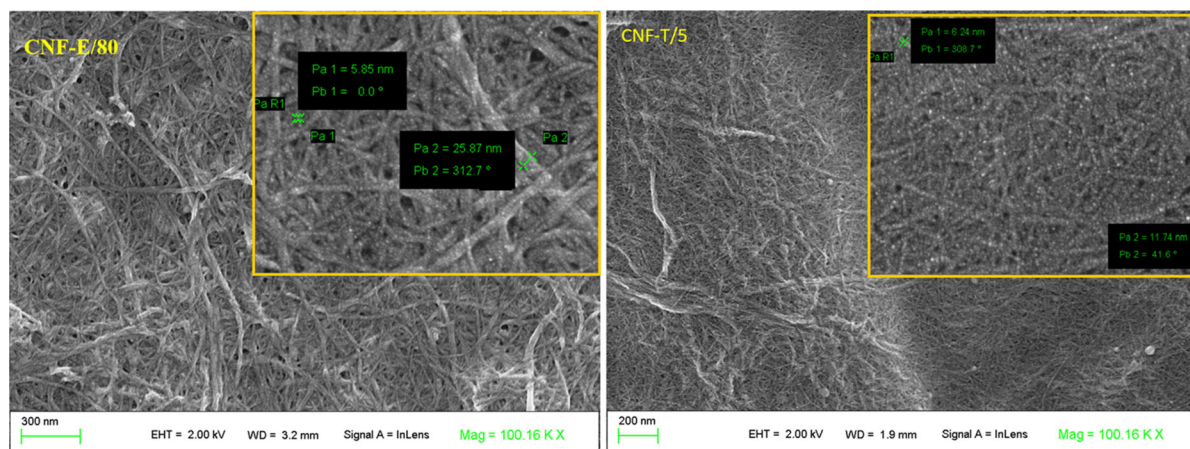


Fig. 4 FE-SEM observation of nanopapers surfaces made of CNF-E (*left*) and CNF-T (*right*) at high magnification

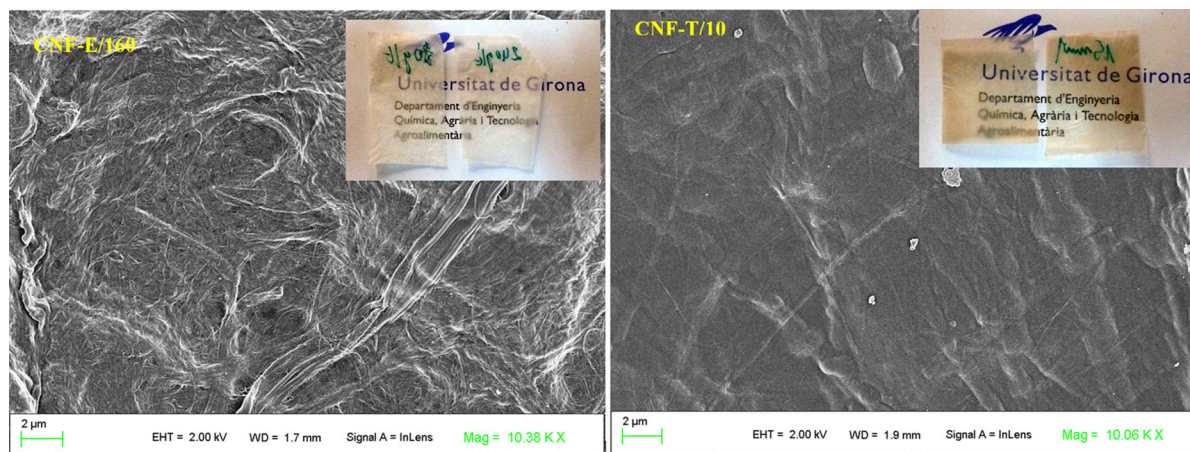


Fig. 5 FE-SEM observation of nanopapers surfaces made of CNF-E (*left*) and CNF-T (*right*) at low magnification

content of CNF-T is apparent again, since it contained 3% more of water at the same environmental conditions. Between 240 and 280 °C, depending on the CNF type, another weight loss can be observed. This stage can be associated to hemicelluloses degradation, which are more sensitive to temperature than cellulose due to its amorphous character and, thus, higher exposition. The presence of carboxyl groups on CNF-T surface clearly affected the thermal stability of such nanopapers, occurring 40 °C before than in the case of CNF-E. The decomposition then continued with a third and last step between 400 and 650 °C, where about 40% of the total weigh was lost. This third step is more perceptible in the case of CNF-T than in the case of CNF-E. Depending on the carboxyl content of the CNF-T, this weight loss was observed at 450 °C

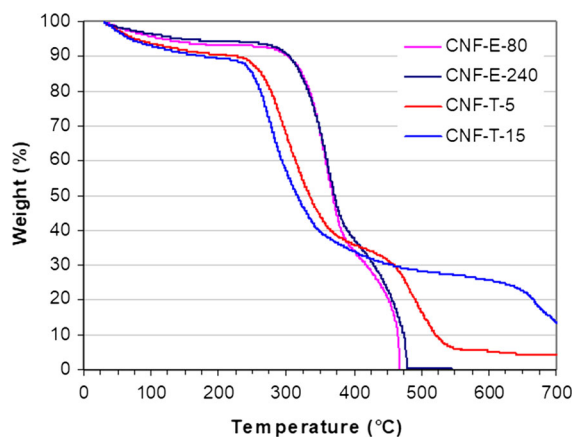


Fig. 6 TGA curves of nanopapers made of CNF-E and CNF-T under oxygen atmosphere

Table 5 Characteristic data from TGA analysis of the CNF samples

Sample	T _{onset} (°C)	Residue at 180 °C (%)	Residues at 500 °C (%)
CNF-T/5	245	10	25
CNF-T/15	240	10.5	30
CNF-E/80	287	7	0
CNF-E/240	285	8	0

(CNF-T/5) or at 650 °C (CNF-T/15). The shift between the temperatures in this third stage denotes the formation of a carbonaceous layer with its thickness depends on the carboxyl content. This layer may delay the thermal decomposition of the organic material and could explain the higher residue found at 600 °C in CNF-T/15.

Conclusions

In the present work, the effect of CNF production method on the final properties of nanopapers has been investigated. CNFs prepared by TEMPO-mediated oxidation showed higher fibrillation yield, transparency and specific surface than that prepared by enzymatic hydrolysis. According to FE-SEM observation, diameters were in the range of 5–10 and 5–25 nm for CNF-T and CNF-E, respectively. These values were in consonance with that calculated using cationic demand and carboxyl content. For the first time, a complete comparative study of nanopapers prepared from CNF-T and CNF-E has been performed, easing the understanding of how the severity of each treatment affects both to CNF and nanopapers.

Nanopapers produced from CNF-E presented lower transmittance than those made of CNF-T. This fact limits the applicability of such nanopapers in products where transparency is a requirement. However, even though the tensile strength and transparency of CNF-T nanopapers was higher, stiffness is limiting as well their use for foldable devices. Density was also found to be a key parameter on the final mechanical performance of nanopapers, especially for tensile strength. Overall, good mechanical properties in both cases were obtained, leading to fully bio-based and biodegradable high-performance structures with high potential of applications in several fields.

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References

- Althues H, Henle J, Kaskel S (2007) Functional inorganic nanofillers for transparent polymers. *Chem Soc Rev* 36(9):1454–1465
- Besbes I, Alila S, Boufi S (2011) Nanofibrillated cellulose from TEMPO-oxidized eucalyptus fibres: effect of the carboxyl content. *Carbohydr Polym* 84(3):975–983
- Chun S-J, Lee S-Y, Doh G-H, Lee S, Kim JH (2011) Preparation of ultrastrong nanopapers using cellulose nanofibrils. *J Ind Eng Chem* 17(3):521–526
- Dankovich TA, Gray DG (2011) Contact angle measurements on smooth nanocrystalline cellulose (I) thin films. *J Adhes Sci Technol* 25(6–7):699–708
- de Marco JL, Felix CR (2007) Purification and characterization of a beta-glucanase produced by *Trichoderma harzianum* showing biocontrol potential. *Braz Arch Biol Technol* 50(1):21–29
- Delgado-Aguilar M, González I, Tarrés Q, Alcalá M, Pèlach MÀ, Mutjé P (2015a) Approaching a low-cost production of cellulose nanofibers for papermaking applications. *BioResources* 10(3):5345–5355
- Delgado-Aguilar M, Tarrés Q, Puig J, Boufi S, Blanco Á, Mutjé P (2015b) Enzymatic refining and cellulose nanofiber addition in papermaking processes from recycled and deinked slurries. *BioResources* 10(3):5730–5743
- Eriksen O, Syverud K, Gregersen O (2008) The use of microfibrillated cellulose produced from kraft pulp as strength enhancer in TMP paper. *Nord Pulp Pap Res J* 23(3):299–304
- Espinosa E, Tarrés Q, Delgado-Aguilar M, González I, Mutjé P, Rodríguez A (2016) Suitability of wheat straw semi-chemical pulp for the fabrication of lignocellulosic nanofibres and their application to papermaking slurries. *Cellulose* 23(1):837–852
- Ferrer A, Quintana E, Filpponen I, Solala I, Vidal T, Rodríguez A, Laine J, Rojas OJ (2012) Effect of residual lignin and heteropolysaccharides in nanofibrillar cellulose and nanopaper from wood fibers. *Cellulose* 19(6):2179–2193
- Henriksson M, Henriksson G, Berglund L, Lindström T (2007) An environmentally friendly method for enzyme-assisted preparation of microfibrillated cellulose (MFC) nanofibers. *Eur Polym J* 43(8):3434–3441
- Henriksson M, Berglund LA, Isaksson P, Lindstrom T, Nishino T (2008) Cellulose nanopaper structures of high toughness. *Biomacromol* 9(6):1579–1585
- Hu L, Zheng G, Yao J, Liu N, Weil B, Eskilsson M, Karabulut E, Ruan Z, Fan S, Bloking JT (2013) Transparent and conductive paper from nanocellulose fibers. *Energy Environ Sci* 6(2):513–518

- Hubbe MA (2014) Prospects for maintaining strength of paper and paperboard products while using less forest resources: a review. *BioResources* 1(9):1787–1823
- Isogai A, Saito T, Fukuzumi H (2011) TEMPO-oxidized cellulose nanofibers. *Nanoscale* 3(1):71–85
- Jorfi M, Foster EJ (2015) Recent advances in nanocellulose for biomedical applications. *J Appl Polym Sci* 132(14). doi:10.1002/app.41719
- Lizundia E, Delgado-Aguilar M, Mutjé P, Fernández E, Robles-Hernandez B, de la Fuente MR, Vilas JL, León LM (2016) Cu-coated cellulose nanopaper for green and low-cost electronics. *Cellulose* 23(3):1997–2010
- Naderi A (2017) Nanofibrillated cellulose: properties reinvestigated. *Cellulose* 24(5):1933–1945
- Naderi A, Lindström T, Pettersson T (2014) The state of carboxymethylated nanofibrils after homogenization-aided dilution from concentrated suspensions: a rheological perspective. *Cellulose* 21(4):2357–2368
- Olsson RT, Samir MA, Salazar-Alvarez G, Belova L, Ström V, Berglund LA, Ikkala O, Nogues J, Gedde UW (2010) Making flexible magnetic aerogels and stiff magnetic nanopaper using cellulose nanofibrils as templates. *Nat Nanotechnol* 5(8):584–588
- Österberg M, Vartiainen J, Lucenius J, Hippi U, Seppälä J, Serimaa R, Laine J (2013) A fast method to produce strong NFC films as a platform for barrier and functional materials. *ACS Appl Mater Interfaces* 5(11):4640–4647
- Saito T, Kimura S, Nishiyama Y, Isogai A (2007) Cellulose nanofibers prepared by TEMPO-mediated oxidation of native cellulose. *Biomacromolecules* 8(8):2485–2491
- Segal L, Creely J, Martin A, Conrad C (1959) An empirical method for estimating the degree of crystallinity of native cellulose using the X-ray diffractometer. *Text Res J* 29(10):786–794
- Sehaqui H, Liu A, Zhou Q, Berglund LA (2010) Fast preparation procedure for large, flat cellulose and cellulose/inorganic nanopaper structures. *Biomacromolecules* 11(9):2195–2198
- Sehaqui H, Zhou Q, Ikkala O, Berglund LA (2011) Strong and tough cellulose nanopaper with high specific surface area and porosity. *Biomacromolecules* 12(10):3638–3644
- Shinoda R, Saito T, Okita Y, Isogai A (2012) Relationship between length and degree of polymerization of TEMPO-oxidized cellulose nanofibrils. *Biomacromolecules* 13(3):842–849
- Spence K, Habibi Y, Dufresne A (2011) Nanocellulose-based composites. *Cellulose fibers: bio-and nano-polymer composites*. Springer, Berlin, pp 179–213
- Sun X, Wu Q, Ren S, Lei T (2015) Comparison of highly transparent all-cellulose nanopaper prepared using sulfuric acid and TEMPO-mediated oxidation methods. *Cellulose* 22(2):1123–1133
- Tarrés Q, Saguer E, Pèlach M, Alcalà M, Delgado-Aguilar M, Mutjé P (2016) The feasibility of incorporating cellulose micro/nanofibers in papermaking processes: the relevance of enzymatic hydrolysis. *Cellulose* 23(2):1433–1445
- Wu C-N, Saito T, Fujisawa S, Fukuzumi H, Isogai A (2012) Ultrastrong and high gas-barrier nanocellulose/clay-layered composites. *Biomacromolecules* 13(6):1927–1932
- Zhu H, Fang Z, Preston C, Li Y, Hu L (2014) Transparent paper: fabrications, properties, and device applications. *Energy Environ Sci* 7(1):269–287